



Habib University

EE-437 Renewable energy systems – Fall 2025

Time = 15 minutes

Quiz 05 – CLO 3

Max Points: 20

CLO 3 - 20 marks

Question 1 [2 points]: How can energy storage help in decoupling temporal and spatial decoupling of generation and consumption of energy?

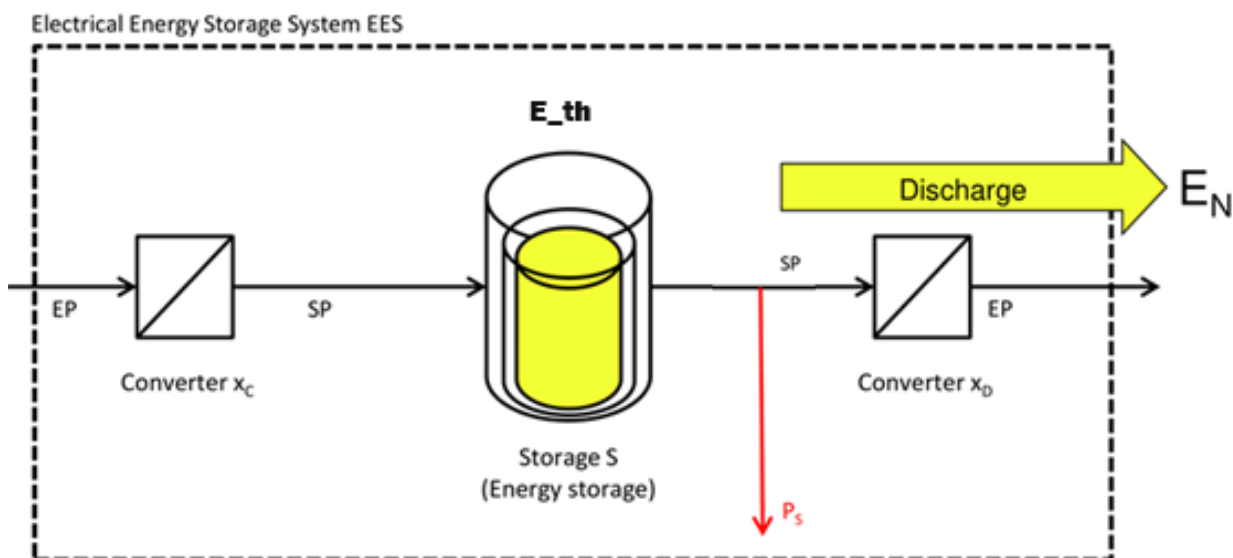
Temporal: Storing energy produced in the day and using at night (time based)

Spatial: Storing energy as fuel and transporting to where its needed (location based)

Question 2 [1 point]: Define peak shaving.

Reducing the peak load in the grid by storage or other means

Question 3 [3 points]: Draw the abstract energy model, labelling x_c , x_d and E_{th} and $E_{nominal}$ and self-discharge.



Question 4 [1 point]: If a battery has 0.5 SoC, what is its DoD? **0.5**

Question 5 [2 points]: Flywheels are used to even out frequencies on the grid level. Why are flywheels more suitable for this operation compared to something like lithium-ion batteries?

Grid stabilization requires high power for very short pulses. Li-Ion would have severe ageing issues. (Any 1 reason is enough)

Question 6 [4 points]: For the following sources match the characteristics to the storage type (redox flow, lithium ion, compressed air, flywheel, supercapacitor, NiMH, pumped hydro, DLC):

- 1- 300 times more power than energy **Flywheel**
- 2- 10 times more power than energy **DLC/Supercapacitor**
- 3- Stores energy as pressure **CAES**
- 4- Power and energy can be sized separately **Redox flow/ flywheel/ PH/ CAES (Any 1)**
- 5- Highest self-discharge **Flywheel**
- 6- Stores energy as gravitational potential energy **Pumped hydro**
- 7- Electric cell with nominal voltage less than 3V **NiMH**
- 8- Needs heat exchangers to work efficiently **CAES**

Question 7 [1 point]: What elements exist in an LFP and MNC cell? Example (Elements in LCO cell are lithium, cobalt and oxygen)

LFP: Lithium, Iron, Phosphorous, Oxygen (any 2)

MNC: Lithium, Manganese, Nickel, Cobalt (any 3)

Question 8 [3 point]: List any 3 reasons a BMS is needed for lithium-ion battery packs.

Sensing

Protection

Interface

Performance management

SOC and SOH estimation

Cell balancing (Any 3)

Question 9 [1 point]: List any 2 effects of ageing on a lithium-ion cell.

Higher impedance,

Higher self-discharge,

Lower capacity (Any 2)

Question 10 [2 points]: Given that for a capacitor, $I = C \frac{dV}{dt}$ and $P = VI$ and $E = \frac{1}{2} CV^2$ derive the energy stored in a capacitor.

$$E = \int P dt = \int VI dt = \int V * C \frac{dV}{dt} dt$$
$$E = C \int V dV = C * \frac{V^2}{2}$$